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Toward Advancing Women's Health in Least Developed Countries: Evaluating Contraceptive Distribution Models in Senegal

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Abstract. *Problem definition:* Improving access to contraceptives is one of the most effective interventions to prevent unintended pregnancies and save the lives of women in least developed countries (LDCs), where the overwhelming majority of maternal deaths occur. However, access to reproductive health commodities is often limited in LDCs because of frequent stock-outs at last-mile health facilities. In this study, we evaluate and compare the effect of two distribution models on last-mile contraceptive availability and key public health outcomes (e.g., unintended pregnancies, maternal and newborn deaths). These distribution models are (i) the commonly used pull distribution model, in which health facilities are fully responsible for managing inventory, and (ii) an alternative model known as the informed push distribution model, which delegates inventory management tasks to external logistics providers. *Methodology/results:* We leverage the staggered transition from pull distribution to informed push distribution in Senegal, a country that redesigned its contraceptive distribution system. We conduct empirical analyses, including a triple differences estimation, on novel field data compiled from multiple sources to evaluate the effect of the transition. We find that the transition significantly reduces contraceptive stock-outs, frontline health worker workload, unintended pregnancies, and maternal and newborn mortalities and also improves client satisfaction, especially in health facilities with less mature inventory management practices and less developed road infrastructure. A comprehensive cost-benefit analysis shows that the aforementioned benefits are achieved in a cost-efficient manner at these facilities, making them prime candidates for the transition. However, for facilities with less mature inventory management practices but more developed road infrastructure, upgrading the inventory management system is a substantially more cost-efficient alternative than transitioning to a new distribution model without compromising the benefits. *Managerial implications:* Given the resource constraints faced by the public health sector in LDCs, it is imperative to understand how the operational and public health benefits of the transition to the informed push model vary based on facility characteristics. Our findings offer actionable insights for resource allocation by identifying health facilities that benefit the most from the transition.

Supplemental Material: The online appendix is available at <https://doi.org/10.1287/msom.2021.0488>.

Keywords: public health supply chains • last-mile distribution • frontline health workers • low-income countries • United Nations Sustainable Development Goals

Across the region, a neglected crisis is silently raging. Unintended pregnancies propel millions of women and girls to drop out of school and workforce, with a lifelong impact on their earnings, health, and welfare.—Dr. Ban-net Ndyanabangi, regional director (United Nations Population Fund 2022)

1. Introduction

Access to reproductive health commodities is of paramount importance, not only for women to exercise their

reproductive rights and avert unintended pregnancies, but also to prevent a spectrum of adverse societal and health outcomes (Plambeck and Ramdas 2020, Alban et al. 2022). These repercussions span from lost career aspirations to mental health concerns (e.g., distress, depression) and physical health issues (e.g., injuries, infections, and even deaths). Despite being recognized as a life-saving health commodity with multifaceted benefits (PATH 2015), access to reproductive health commodities is often limited in the least developed countries (LDCs), defined

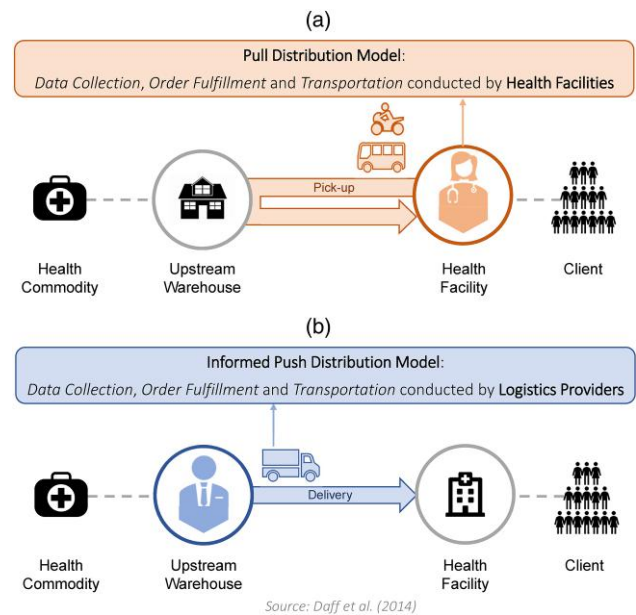
as “low-income countries confronting severe structural impediments to sustainable development” (United Nations 2023). For example, the rate of unmet need for contraceptives¹ is estimated to be around 50% in LDCs. This is in stark contrast to the approximately 14% rate in upper middle income countries and the 6% unmet need in high-income countries (Darroch 2017).

The limited access to reproductive health commodities in LDCs is driven by both demand-side (e.g., public awareness) and supply-side factors. On the supply side, a critical barrier to access is the stock-outs of health commodities at last-mile facilities (Gallien et al. 2021, Karamshetty et al. 2022). Evidence from field surveys suggests that such stock-outs are widespread across LDCs from Latin America to Sub-Saharan Africa and South Asia (Gallien et al. 2017, Karimi et al. 2021). Within this context, an important factor that affects the last-mile availability of health commodities is the distribution model used to deliver such commodities from upstream warehouses to downstream health facilities. Of note, the most commonly used distribution model in LDCs is the pull distribution model (Yadav et al. 2011), in which frontline health workers at last-mile facilities are responsible for executing the activities of the logistics cycle (U.S. Agency for International Development 2011b): data collection (i.e., tracking and recording of inventory data), order fulfillment (i.e., transmitting resupply orders to upstream locations), and transportation (i.e., transporting supplies from upstream locations to last-mile facilities (see Figure 1(a)). A notable drawback of the pull distribution model is that the frontline health workers, whose primary roles are to provide clinical services, are not equipped with the necessary skills and resources to effectively execute the activities of the logistics cycle. Further, managing clinical and nonclinical tasks results in heavy workload for the health workers, compromising the quality and effectiveness of both activities. This could result in stock-outs and diminished patient satisfaction.

To mitigate these challenges, several LDCs have adopted an alternative distribution approach known as the informed push distribution model (see Daff et al. 2014). This model utilizes trained logistics providers who make scheduled visits to each last-mile health facility to carry out the activities of the logistics cycle in lieu of frontline health workers performing those tasks (U.S. Agency for International Development 2011b, Daff et al. 2014) (see Figure 1(b)). Specifically, logistics providers are responsible for the last-mile transportation and delivery of health commodities. In addition, during delivery visits, they verify the accuracy of data collected by the frontline health workers and subsequently record and transmit it upstream.

Although a transition to the informed push model has potential benefits—as evidenced by anecdotal and descriptive insights (Daff et al. 2014, Mbow et al. 2017)—it is capital-intensive requiring substantial start-up

Figure 1. (Color online) Public Health Supply Chains in Least Developed Countries



Note. (a) Pull distribution versus (b) informed push distribution.

investments (e.g., to fund the purchase of delivery vehicles, hiring and training logistics providers) as well as high operating costs (e.g., for fuel, vehicle maintenance, and wages for logistics providers).² Hence, it is imperative for policymakers to ensure that the resources allocated toward transitioning to the informed push model yield operational and public health benefits that justify the substantial costs. This consideration is particularly relevant in the context of LDCs, where the availability of funding and other resources is often severely constrained, motivating the need for a rigorous evaluation of the impact of the transition to the informed push model. As noted by Oxman et al. (2010, p. 427), “both politically, in terms of being accountable to those who fund the system, and also ethically, in terms of making sure that you make the best use possible of available resources, evaluation is absolutely critical.” However, a review of the prior literature—on public health supply chains in LDCs (e.g., Parvin et al. 2018, Gallien et al. 2021, Karimi et al. 2021) and on the performance evaluation of distribution models for health commodities (e.g., Daff et al. 2014, Krug et al. 2020)—provides a limited understanding of the operational, public health, and economic implications of the transition to an informed push model. Consequently, we investigate the following research questions (RQs) in our study:

RQ1. Operational implications: Under what conditions is the transition from a pull model to an informed push model more effective in reducing contraceptive stock-outs in public health supply chains?

RQ2. Public health implications: What is the impact of the transition on key public health outcomes,

including client satisfaction, contraceptive prevalence rate, unintended pregnancies, and maternal and newborn mortalities?³

RQ3. Economic implications: Given the cost–benefit trade-offs associated with the transition and the severe resource constraints faced by the public health sector in LDCs, which facilities should be prioritized for the transition?

Our first research question (RQ1) is aimed at evaluating contingencies that may amplify or attenuate the effects of the transition from the pull model to the informed push model in reducing contraceptive stock-outs. We examine contingencies at the health facility level that are tightly intertwined with the execution of the logistics cycle activities, namely, a health facility's logistics management information system (LMIS) practices (a system for tracking and reporting inventory data) and road infrastructure access. We refer to a health facility's LMIS practices as more mature when the facility engages in either the frequent updating of inventory records or the use of electronic LMIS or both and as less mature when the facility engages in neither of those practices. A health facility's access to road infrastructure is referred to as more developed when it is in proximity to a primary road (i.e., lower than the median distance in our sample) and less developed otherwise. Based on these contingencies, we classify health facilities as severely disadvantaged when they are low in both moderating variables (i.e., less mature LMIS practices and less developed road infrastructure), as moderately disadvantaged when they are low in one variable only, and as nondisadvantaged when they are high in both variables. Our second research question (RQ2) is aimed at evaluating the public health implications of the transition. Addressing this research question is crucial because the ultimate objective of supply chain interventions such as the informed push model is to improve the health and well-being of populations. Finally, our third research question (RQ3) is focused on conducting a comprehensive evaluation of the economic implications of the transition to the informed push model. This assessment is instrumental in identifying the health facilities that should be prioritized for the transition.

To address these research questions, we focus on a public health supply chain intervention in Senegal, West Africa, where the model used for distributing contraceptive methods to last-mile facilities transitioned from the pull model to the informed push model in a staggered manner (Daff et al. 2014). Prior to this transition, Senegal was among the lowest ranked countries in the world in terms of reproductive health commodity usage (World Bank 2011), where at least one third of women who wanted to avoid a pregnancy were not using modern contraception (Cavallaro et al. 2016). This posed a substantial barrier to the country's progress toward the United Nations Sustainable Development Goals (UN

SDGs). The supply chain intervention was aimed at mitigating this barrier through expanded contraceptive access. As such, evaluating its impact holds practical relevance for improving the accessibility of reproductive health commodities in Senegal and LDCs more broadly.

The staggered rollout of the informed push model allows us to employ a difference-in-differences (DiD) estimation, in which never-treated health facilities and those that are yet to be treated serve as control units for health facilities that are already treated. However, this estimation may lead to biased results if the two groups of facilities are inherently different from one another or if the treatment group is impacted by unobserved confounding variables in the post-treatment period. Hence, to evaluate the first order effects of the informed push model implementation, we add a third difference to the DiD model by incorporating deworming medications as placebo units that were unaffected by the transition. This approach, known as a triple differences estimation method, alleviates bias from facility-level unobserved confounding variables.

Our empirical analysis yields several actionable insights as described subsequently. First, regarding the operational implications (RQ1), we find that the transition to the informed push model is most effective in reducing stock-outs at severely disadvantaged health facilities, leading to a 30.00 percentage-point (p.p.) reduction in stock-outs. However, the benefits are significantly lower in the case of moderately disadvantaged health facilities and those that are nondisadvantaged. Second, the public health implications (RQ2) of the transition are also on similar lines. That is, severely disadvantaged health facilities stand to gain the most from the transition in terms of reducing frontline health worker workload and improving key client outcomes. These outcomes include enhanced client satisfaction, increase in contraceptive prevalence rate (defined as the percentage of women or their partners currently using a contraceptive method), and reductions in unintended pregnancies as well as maternal and newborn mortalities. Finally, with respect to the economic implications (RQ3), the results of a comprehensive cost–benefit analysis indicate that the transition to the informed push model is not only most effective in improving various outcomes at the severely disadvantaged health facilities but also most cost-efficient.

Whereas our study's primary focus is evaluating the impact of the transition from the pull model to the informed push model, there are alternative initiatives that can potentially improve the availability of health commodities. One such initiative is investing to enhance the maturity of a facility's LMIS practices. Within this context, we find that, for severely disadvantaged health facilities, the transition to the informed push model may cost more, but it also delivers substantially larger benefits compared with investments in upgrading a facility's

LMIS practices. Hence, when funding constraints are not too tight, it would be prudent to transition such facilities to the informed push model. In contrast, for facilities that are disadvantaged only with respect to LMIS practices, upgrading the LMIS practices is a substantially more cost-efficient strategy compared with a transition to the informed push model. Collectively, our research provides actionable insights for public health supply chain managers and policymakers in LDCs by outlining strategies for the effective and efficient use of limited resources to increase access to reproductive health commodities.

2. Literature Review

2.1. Managing Public Health Supply Chains in LDCs

The first research stream related to our paper is the operations management literature on public health supply chains in LDCs, settings that are often characterized by scarce budgets, lack of access to roads, and poor data management practices (see Pedraza-Martinez and Van Wassenhove 2016, Jónasson et al. 2017). In this stream, several papers study the impact of factors that influence the availability of health commodities at a macrolevel (e.g., national level). For instance, Kazaz et al. (2016) study the effect of different policy interventions aimed at increasing the availability of artemisinin-based malaria medicine in Sub-Saharan Africa and find that a minimum support price offered by the government improves product availability. In addition, several papers investigate the relationship between funding and health commodity availability at the national level. For example, Gallien et al. (2017) use simulation models to analyze the impact of procurement and grant disbursement process on drug availability across Africa. Natarajan and Swaminathan (2014) find that funding delays negatively impact product availability and operating costs in humanitarian health supply chains.

Beyond the national level, several papers in this stream examine factors that impact the availability of health commodities at last-mile health facilities. Focusing on the Zambian public health supply chain, Gallien et al. (2021) propose optimal inventory control policies to reduce the last-mile stock-outs of antimalarial medications. Karamshetty et al. (2022) conduct semistructured interviews with managers of health facilities in Nairobi to develop a qualitative understanding of the drivers of underperformance (e.g., stock-outs). Parvin et al. (2018) develop analytical models to compare the effectiveness of two distribution strategies in mitigating stock-outs of highly seasonal antimalarial medications. They find that, despite being less cost-efficient, a delayed shipment strategy is more effective at reducing shortages compared with a transshipment model. De Boeck et al. (2022) study vaccine distribution in LDCs when faced with logistical challenges during rainy seasons.

Our paper is distinguished from the cited literature in three main ways. First, prior research studies the availability of health commodities at either the national level or last-mile health facilities in isolation. However, except for a few studies (e.g., Parvin et al. 2018, De Boeck et al. 2022), there is limited investigation into the impact of product distribution models that link the national level and last-mile health facilities. Of note, we found no studies that compare the effectiveness of the commonly prevalent pull distribution model to that of the informed push model in reducing stock-outs. Investigating this relationship is important because the national-level availability of supplies does not directly translate into last-mile availability. For instance, Leung et al. (2016, p. 1) report the prevalence of stock-outs at the last-mile level “despite ample inventory being simultaneously available at the national warehouse.” Second, unlike previous research, we evaluate the effect of product distribution models not only on health commodity stock-outs but also on public health outcomes. As Starr and Van Wassenhove (2014, p. 930) point out, adapting operations management methods to “capture the multitude of stakeholders as well as their behaviors is a vital avenue for research,” particularly in the context of public health supply chains in LDCs. Third, most papers in this stream are based on analytical or simulation models. This can be primarily attributed to the challenges of collecting large-scale, granular, and reliable data at the health facility level in LDCs. We attempt to overcome these challenges by analyzing novel field data collected across multiple entities, including health facilities, frontline health workers, and end clients in Senegal, West Africa.

2.2. Evaluating the Performance of Pull vs. Informed Push Distribution Models

The second literature stream related to our paper focuses on evaluating the performance of pull versus informed push distribution in LDCs. In this stream, several practitioner-oriented reports and white papers highlight the challenges of last-mile delivery with the pull distribution model. For instance, the U.S. Agency for International Development (2011b, p. 1) (USAID) argues that the last mile represents “a critical challenge in ensuring access to health commodities” under the pull model. Similarly, VillageReach (2017) highlights the challenges faced by health facilities with the pull model in which frontline health workers often take time away from clinical services to place orders and pick up supplies from upstream locations. Yadav et al. (2011, p. 7) provide an overarching review of public health supply chains in such settings and argue that the informed push model is robust to “weak order and stock management capabilities at the lowest level of the distribution system.”

In addition, several studies compare the performance of the pull model against that of the informed push model in Senegal. For example, based on qualitative

interviews, Mbow et al. (2017, p. 30) find that the informed push model improves the availability of health commodities “even in the most peripheral structures such as health huts.” However, they do not report the magnitude of the improvement in product availability. Using descriptive statistics on a sample of 14 health facilities in Senegal, Daff et al. (2014) conclude that a transition to the informed push model reduces the risk of contraceptive stock-outs to less than 2% for the average health facility. Similarly, Hasselback et al. (2017) present descriptive statistics on data collected from health facilities in Senegal that indicate a reduction in contraceptive stock-outs after a transition to the informed push model. Krug et al. (2020) use self-reported facility-level data collected from a subset of regions in Senegal and apply an interrupted time-series analysis without the assignment of effective control units to evaluate the impact of the transition to the informed push model. They claim a reduction in stock-outs following the transition but do not report quantitative estimates of the reductions.

Our paper differs from the cited literature in multiple ways. First, previous studies that evaluate the effect of the informed push model are based largely on qualitative arguments (Mbow et al. 2017), descriptive statistics in small-scale pilot settings (U.S. Agency for International Development 2008, Daff et al. 2014, Hasselback et al. 2017), or analyses that do not account for unobserved heterogeneity in the sample (Krug et al. 2020). As a result, these studies do not provide a robust causal assessment of the impact of the informed push model on stock-outs. Second, we identify key health facility-level infrastructural variables that moderate the effect of a transition to the informed push model. These factors are not explored in prior studies. Third, we compare the cost-efficiency of the informed push model to that of alternative initiatives that could be used to improve the last-mile availability of health commodities. This analysis uncovers nuanced insights for maximizing the effectiveness and efficiency of resource allocation in public health supply chains. Finally, we evaluate how the supply chain benefits of the informed push model translate into improvements in public health outcomes, aspects that are not explored in prior studies.

3. Empirical Setting

With the traditional pull distribution, health facilities are responsible for executing the three activities of the logistic cycle (i.e., data collection, order fulfillment, and transportation), creating significant challenges with respect to the timeliness and predictability of supply refills. For example, Senegal's pull distribution model relied on the frontline health workers to “accurately forecast, track, and order contraceptives” (i.e., to perform data collection-related activities; see Hasselback et al. 2017, p. 2) even though their primary responsibility was clinical

services and they lacked proper training in inventory management (Daff et al. 2014). With respect to order fulfillment and transportation, frontline health workers in Senegal often traveled to upstream warehouses using ad hoc modes of transportation (e.g., public transport, privately owned vehicles), which were not readily available when needed, resulting in uncertainties and delays in replenishment. These challenges reduced visibility into consumption at last-mile health facilities and increased the supply lead times, elevating the risk of last-mile stock-outs.

Given the widespread prevalence of stock-outs with the traditional pull distribution model, the Senegalese Ministry of Health and Social Action partnered with the National Supply Pharmacy, IntraHealth International, and the Bill & Melinda Gates Foundation to develop an alternate distribution strategy named the informed push model. Under the informed push model, the activities of the logistics cycle are delegated to trained logistics providers who are equipped with dedicated and more reliable means of transportation (e.g., pickup trucks). The logistics providers deliver commodities to the last-mile health facilities on a monthly schedule, eliminating the need for facilities to pick up supplies from upstream warehouses. During the delivery visits, logistics providers verify the accuracy of information recorded on stock cards and make necessary adjustments. In addition, they calculate the resupply amounts and top up inventories based on prespecified minimum–maximum levels (U.S. Agency for International Development 2011a, Leung et al. 2016). Finally, logistics providers transmit the verified inventory data to upstream warehouses, enhancing the visibility of last-mile consumption. These activities have the potential to improve data accuracy, enhance visibility, and expedite the replenishment process. Collectively, these improvements can result in increased last-mile availability of health commodities under the informed push model.

In addition to changes in logistics cycle activities, the rollout of the informed push model involved a modification in the payment arrangement. With this change, health facilities were no longer required to make upfront payments for replenished contraceptives from warehouses; instead, payments could be made after product dispensation. Given the concurrent occurrence of these modifications in logistics cycle activities and payment structure, we are limited in our ability to fully disentangle the effects of the two on the outcomes. We acknowledge this limitation and discuss the implications in Section 7.2 of the paper.

3.1. Operational Implications of the Transition: Effect on Contraceptive Stock-outs (RQ1)

Notwithstanding the potential benefits of the transition to the informed push model, one notable drawback of this model is the need for considerable start-up

investments (e.g., to acquire delivery vehicles, conduct training) and operating costs (e.g., for fuel, maintenance, supervision; see Daff et al. 2014). Given the tight resource constraints in LDCs, identifying the conditions under which the transition to the informed push model is likely to be more (versus less) beneficial is of significant relevance to policymakers in these countries. To this end, we focus on two key infrastructural characteristics of health facilities that are intertwined with the execution of the logistics cycle activities: (i) a health facility's LMIS practices for recording and reporting inventory data and (ii) the facility's access to road infrastructure.

3.1.1. Impact of LMIS Practices. We expect that health facilities with less mature LMIS practices (i.e., facilities that neither update their inventory records frequently nor use electronic LMIS) are likely to reap greater benefits from the transition to the informed push model compared with those with more mature LMIS practices (i.e., facilities that engage in either or both of the aforementioned practices). The underlying rationale is as follows. First, the lack of frequent updating of inventory records—which involves logging changes as soon as contraceptives are received from the warehouses or dispensed to clients—can result in less accurate and up-to-date inventory information at health facilities operating under the pull distribution model. Second, the absence of electronic LMIS for inventory management necessitates the manual preparation and delivery of inventory reports by health workers or postal services to upstream locations, leading to inaccuracies and delays in the replenishment process. In contrast, the usage of electronic LMIS allows for a more rapid and accurate compilation and transmission of such data upstream (see U.S. Agency for International Development 2011a). Taken together, facilities that are not engaged in the aforementioned LMIS practices may suffer from decreased accuracy and timeliness of inventory information, increasing the likelihood of supply–demand mismatches and stock-outs. Consequently, we anticipate that health facilities with less mature LMIS practices are likely to derive greater benefits from transitioning to the informed push model than facilities with more mature LMIS practices.

3.1.2. Impact of Road Infrastructure. We expect that health facilities with access to less developed road infrastructure will benefit more from transitioning to the informed push model compared with those with access to more developed road infrastructure. This is because limited access to road infrastructure under the pull distribution model can hinder the effective order fulfillment and transportation of health commodities (Yadav et al. 2011). For instance, with the pull model, health facilities often rely on public transportation to reach upstream warehouses, and access to such modes of transport could be more restricted for facilities in remote and off-road

areas (Kolapo et al. 2009). In addition, health facilities may rely on other ad hoc modes of transport (e.g., ambulance, privately owned vehicle), which may not always be available when needed. Further, one unique aspect of the last mile in LDCs is that “road infrastructure typically worsens as one travels farther away from the point of resupply” (U.S. Agency for International Development 2011b, p. 2). Consequently, health workers at remote facilities may not only commute on roads that are in poorer condition (e.g., dirt roads), but also travel longer distances to access the resupply locations, exacerbating the delays in supply lead times. In sum, health facilities lacking access to adequate road infrastructure face substantial challenges with respect to effective order fulfillment and transportation of supplies. Consequently, these facilities are likely to experience larger reductions in stock-outs when they transition to the informed push model compared with facilities that have access to more developed road infrastructure.

3.2. Public Health Implications of the Transition: Effect on Health Workers and Clients (RQ2)

Beyond the operational implications discussed (RQ1), the transition to the informed push model is also likely to yield public health benefits. Examining such public health implications of the transition is a crucial component of our study (RQ2) given that the overarching objective of supply chain interventions such as the informed push model is to ultimately improve community health and well-being.

3.2.1. Impact on Frontline Health Workers. With the traditional pull distribution model, frontline health workers need to take time away from their primary responsibility of providing clinical services to perform noncore activities such as tracking and ordering health supplies (Hasselback et al. 2017). This added responsibility exacerbates the workload of an already overstretched workforce. The transition to the informed push model could help to reduce the workloads of health workers by delegating nonclinical responsibilities to external logistics providers. By enabling frontline health workers to dedicate more time to clinical services, the transition stands to improve public health outcomes. A reduction in workloads, for example, can reduce the instances of rushed or incomplete delivery of clinical services (e.g., family planning counseling, administration of intravenous medications). This can also shorten wait times and prevent patients from departing the facility without receiving the necessary attention and care.

3.2.2. Impact on Clients. The pervasive stock-outs of health commodities under the traditional pull model are likely to have significant negative consequences for clients. This can manifest as decreased client satisfaction, which can subsequently lead to reduced contraceptive

usage, increased risk of method discontinuation, and diminished confidence in the public healthcare delivery systems (Hutchinson et al. 2011). In addition, stock-outs are directly associated with adverse health outcomes, such as unintended and closely spaced pregnancies and maternal and newborn mortalities. Recognizing these ramifications, we evaluate how the operational implications of the transition to the informed push model (i.e., supply-side benefits in terms of reduction in contraceptive stock-outs) carry over to the client side. Specifically, we evaluate the effect of the transition on the following key client-related outcomes: (i) client satisfaction, (ii) contraceptive prevalence rate, (iii) unintended pregnancies, and (iv) maternal and newborn mortalities.

3.3. Economic Implications of the Transition: Evaluating the Cost–Benefit Trade-offs (RQ3)

In Section 3.1, we discuss the operational implications of the transition to the informed push model and the need to identify health facilities where the transition is likely to be more (versus less) beneficial. However, the precise identification of such priority facilities also requires an understanding of the economic costs of the transition (discussed in detail later in Section 4.4). In addressing RQ3, we combine such economic costs of the transition with the estimated benefits (i.e., stock-out reductions) to develop cost–benefit ratios that measure the cost-efficiency of the transition. Specifically, we evaluate how the cost-efficiency of the transition varies based on the infrastructural characteristics of health facilities, namely, the maturity of their LMIS practices and access to road infrastructure. This allows us to identify the specific types of health facilities that should receive priority in resource allocation when transitioning to the informed push model.

4. Research Design

4.1. Data Sources

To address the three research questions, we analyze field data collected by the Demographic and Health Surveys (DHS) program. Funded by USAID, the DHS program collects nationally representative data on various indicators that track countries' progress toward the UN SDGs. These data are routinely used by Ministries of Health and other organizations involved in managing public health supply chains (e.g., USAID, World Health Organization) to inform key operational and policy decisions. To evaluate the operational and public health implications of the transition to the informed push model, we leverage two sources of data collected by the DHS program: (i) service provision assessment (SPA) surveys and (ii) DHS surveys.

Within the SPA surveys, the inventory questionnaire is targeted at health facilities to assess their readiness to deliver essential health services (e.g., family planning)

and commodities (e.g., contraceptives). The surveys are pretested to detect potential issues before they are administered in the field using mobile devices. In addition, survey administrators go through multiple rounds of rigorous training involving classroom lectures/discussions, mock interviews, and role play. Data collected from health facilities are then used to develop a comprehensive picture of a country's capacity to deliver quality health services and products. During survey implementation, a team of interviewers visited each facility and collected data by engaging the "most knowledgeable" person available for the relevant health service (Ministry of Health Malawi and ICF International 2014). We utilized the SPA's inventory questionnaire data collected during the survey rounds of 2012–2013,⁴ 2014, and 2015 to capture stock-outs. These data sources are also used to measure the maturity of LMIS practices at health facilities in Senegal. Of note, health facilities are chosen to be included in the SPA surveys based on a stratified random sampling approach. Consequently, a facility's probability of being selected in a survey does not depend on its selection status in the previous rounds. Whereas this approach ensures that the samples are nationally representative, the survey is not designed to track the same health facilities over time across different rounds. In other words, our data are repeated cross-sectional in nature.⁵

Another component of the SPA surveys is the health worker interview questionnaire, which is designed to collect various information at the frontline health worker level. In Senegal, the survey administrators identified and interviewed up to eight health workers at each facility on the day of the survey visit. We utilize the SPA's health worker interview questionnaire from the survey rounds of 2012–2013, 2014, and 2015 to measure the workload of frontline health workers. Further, to evaluate the impact of the informed push model on client satisfaction, we utilize the SPA's client exit interview questionnaire data to assess the clients' perception of health service delivery. During these interviews, clients are asked whether they perceive specific aspects of the service delivery (e.g., cleanliness of the facility, availability of contraceptives, etc.) to be a major problem, a minor problem, or no problem at all. Using a random sampling approach, the SPA survey team identified and interviewed up to five clients per family planning provider and up to 15 clients at a given health facility (Ministry of Health Malawi and ICF International 2014). Therefore, the clients' perception of service delivery is expected to represent the overall client experience on the day of the exit interview. We collected the SPA's client exit interview data from the survey rounds of 2012–2013 and 2015 to capture client satisfaction.⁶

Finally, the DHS surveys are used to collect nationally representative, repeated cross-sectional data on the demographic and socioeconomic characteristics of the

populations in LDCs. Within the DHS surveys, the women questionnaire is targeted at women of reproductive age (i.e., 15–49 years old) and captures various indicators including age, education, fertility, and mortality. Another questionnaire that is part of the DHS survey is the household questionnaire, which captures additional indicators at the household level, including household size, wealth, and clean water access. In our study, we use the DHS's women and household questionnaire data from the survey rounds of 2012–2013, 2014, and 2015 to capture key demographic and socioeconomic characteristics of the catchment population surrounding a health facility (e.g., age, education).⁷ We supplement these data sources with road network data collected from the OpenStreetMap database to construct our measure of access to road infrastructure.

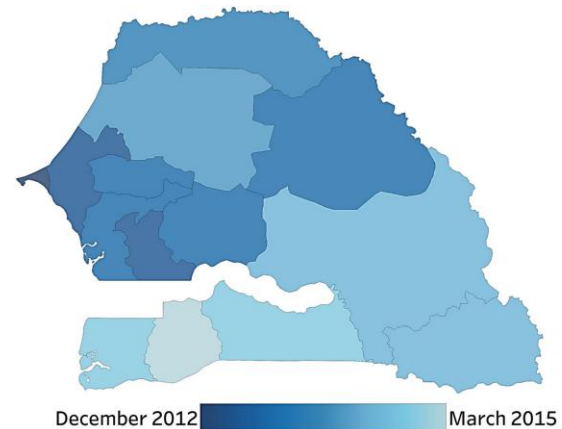
4.2. Evaluating the Operational Implications of the Transition to the Informed Push Model (RQ1)

4.2.1. Variables

4.2.1.1. Dependent Variable. The main dependent variable of interest in this study is *Stock-Out*, a binary variable that indicates whether a particular health commodity was out of stock on the day of the survey visit. Survey administrators verify the availability of health commodities in an objective manner by directly observing them on the shelf. This mitigates concerns related to self-reported information and enhances data reliability. We coded *Stock-Out* as zero when survey administrators noted “at least one valid item was observed” or “reported as available, but not observed” (e.g., because of the storage area being inaccessible)⁸ and as one when survey administrators noted “no valid items observed” or “not available today.” Commodities reported to be “never available” were dropped from the sample as the health facility never offered them (Ministry of Health Malawi and ICF International 2014, p. 218).

4.2.1.2. Independent Variable. The independent variable of interest is *Informed Push Model*, a binary variable that captures the staggered rollout of the transition across regions in Senegal (see Figure 2 and Online Tables A.1 and A.2 for details regarding the staggered rollout).⁹ We assign this variable the value of one after the rollout of the informed push model to public health facilities in a region and zero otherwise. We also assign this variable the value of zero for private health facilities at which the informed push model was never rolled out. We consider all public health facilities in a region as treated after the informed push model is rolled out to that region. In addition, we apply a two-month lag after the starting rollout date to allow the informed push model to become fully operational in each region and omit observations during this window to prevent treatment contamination issues.¹⁰ We note that the full-scale rollout of the informed

Figure 2. (Color online) Staggered Rollout of the Informed Push Model Across Senegal



Source: Cavallaro et al. (2016).

push model across Senegal was preceded by a small pilot phase. During the pilot phase, the informed push model was pretested in two districts (i.e., Pikine and Kaolack districts in Dakar and Kaolack regions). These regions were among the first where the model was scaled up, followed by rollout to the rest of the regions in a staggered manner. To account for potential endogeneity issues arising from the staggered rollout of the transition, we apply several adjustments to our main model as discussed in Section 4.2.2.

4.2.1.3. Moderating Variables. We have two moderating variables, a facility's LMIS practices and access to road infrastructure. We capture a facility's *LMIS Practices* using two metrics: (i) LMIS updating frequency, which measures whether facilities update their inventory records daily, and (ii) electronic LMIS, which measures whether facilities utilize an electronic LMIS (versus paper-based LMIS) to manage and report inventory data. We classify a facility's LMIS practices as less mature when the facility neither updates LMIS daily nor uses electronic LMIS and as more mature when at least one of them is true.

Further, we measure *Road Infrastructure* as the Euclidean distance (in kilometers) between each facility and its nearest primary road, for which smaller values of distance represent better access to road infrastructure. We use the median value of distance to assign facilities into more developed (i.e., lower than the median value) or less developed road infrastructure (i.e., higher than the median value) groups. The choice of distance to road to capture access to road infrastructure is well-grounded in prior literature (e.g., U.S. Agency for International Development 2011b). To construct this measure, we obtained the Senegalese road network data from OpenStreetMap and paired these data with the GPS coordinates of health facilities collected from the SPA surveys.

4.2.1.4. Control Variables. We control for multiple health commodity-, facility-, and client-related factors that can impact the likelihood of stock-outs. Online Tables A.3 and A.4a provide descriptions of the variables used in our model and their summary statistics, respectively.

4.2.2. Model Specification. We apply a triple differences estimation to evaluate the effect of the informed push model (versus pull model) on the main dependent variable: contraceptive stock-outs. Specifically, we estimate a logistic regression based on the following specification (Cunningham 2021), in which ε_{ijrt} is the error term; i, j, r , and t denote health commodity i at facility j located in region r at time t ; and X_{CL} is the vector of control variables:

$$\begin{aligned} & \text{Ln} \left[\frac{\text{Pr}(\text{Stock-Out}_{ijrt} = 1)}{1 - \text{Pr}(\text{Stock-Out}_{ijrt} = 1)} \right] \\ &= \beta_0 + \lambda.X_{CL} + \text{Catch.Pop.Proxy}_{rt} + \alpha.\text{Region}_r + \gamma.\text{Time}_t \\ &+ \beta_1.\text{Informed Push Model}_{rt} \times \text{Contraceptive}_i + \varepsilon_{ijrt}. \end{aligned} \quad (1)$$

In this equation, *Region* represents region fixed effects and *Time* is the vector of year and month fixed effects. *Informed Push Model* is the staggered rollout of the transition across regions in Senegal, estimating a DiD coefficient. In this DiD estimation, the treatment group is composed of public health facilities that have already received the informed push model, whereas the control group includes either public facilities awaiting treatment or private facilities. However, this estimation may yield biased results if the treatment and control units differ inherently from each other or if the treatment group is affected by facility-level unobserved confounding variables during the post-treatment period. To alleviate such endogeneity bias, we add a third difference to the DiD model by incorporating the stock-outs of a health commodity other than contraceptives, namely, deworming medications, as placebo units. The deworming medications serve as appropriate placebo units because the rollout of the informed push model did not involve the distribution of these medications during the study time frame.¹¹ This approach, commonly known as the triple differences estimation method, captures the difference between two DiD estimations, one for units that were affected by the treatment and another for those unaffected (see Cunningham 2021). Hence, in the preceding equation, the variable *Contraceptive* is assigned the value of one for contraceptives¹² and zero for deworming medications.¹³

Further, we utilize coarsened exact matching (CEM) to match treated (i.e., public) and never-treated health facilities (i.e., private) to alleviate concerns of diverging trends in the pretreatment period. Of note, CEM is known to outperform alternative matching approaches (e.g., propensity score matching) in terms of its ability to “reduce

imbalance, model dependence, estimation error, bias, variance, mean square error, and other criteria” (Iacus et al. 2012, p. 2). The balance diagnostics before and after the application of CEM are presented in Online Table B.1a.¹⁴

Finally, our specification controls for a proxy variable to capture the potential endogeneity of rollout timing. Specifically, we identify catchment population size as an important unobserved variable (η) that determines the rollout timing of the informed push model. That is, health facilities located in more populated areas were more likely to be transitioned to the informed push model at an earlier time compared with those located in less populated areas. To address endogeneity issues arising from such an unobserved variable, Freyaldenhoven et al. (2019) suggest finding a variable x that serves as a reasonable proxy for η . They argue that including this proxy variable in the main model corrects for potential endogeneity issues if x is an accurate proxy for the unobserved variable η . In our setting, we identify the overall population of regions in Senegal as a reasonable proxy (x) for the unobserved variable (η). We include this proxy variable, that is, *Catch. Pop. Proxy*, as a control variable in Equation (1). Next, we apply the specification to subgroups of health facilities categorized based on *LMIS Practices* and *Road Infrastructure*, allowing us to compare the benefits of the transition across different subgroups of health facilities.

4.3. Evaluating the Public Health Implications of the Transition to the Informed Push Model (RQ2)

4.3.1. Variables

4.3.1.1. Dependent Variables. We focus on two dependent variables to address RQ2: *Frontline Health Worker Workload* and *Client Satisfaction*. The former is a binary variable that takes the value of one when a worker expressed concern during questionnaire interviews that high “workload” was a factor hindering their “ability to provide good quality of care services” (Ministry of Health Malawi and ICF International 2014, p. 283) and zero otherwise. Similarly, *Client Satisfaction* is a binary variable that takes the value of one when clients reported no major or minor complaints regarding the availability of contraceptive methods and zero otherwise.

4.3.1.2. Independent Variable. The independent variable is the *Informed Push Model*, which is a binary variable that indicates the staggered transition to the informed push model (see Section 4.2.1 for more details).

4.3.1.3. Moderating Variables. The moderating variables in this analysis are a health facility’s (i) *LMIS Practices* and (ii) access to *Road Infrastructure* (see Section 4.2.1 for more details).

4.3.1.4. Control Variables. We control for multiple facility- and client-related factors when evaluating the public health implications of the transition on the two dependent variables. For details related to estimation of models with *Frontline Health Worker Workload* as the dependent variable, see Online Tables A.4b and B.2, and for details related to models with *Client Satisfaction* as the dependent variable, see Online Tables A.4c and B.3.

4.3.2. Model Specification. We apply DiD estimation to evaluate the effect of the informed push (versus pull) model on *Frontline Health Worker Workload* as specified:

$$\begin{aligned} & \text{Ln} \left[\frac{\Pr(\text{Frontline Health Worker Workload}_{hjt} = 1)}{1 - \Pr(\text{Frontline Health Worker Workload}_{hjt} = 1)} \right] \\ &= \beta_0 + \lambda.X_{CL} + \text{Catch.Pop.Proxy}_{rt} + \alpha.Region_r + \gamma.Time_t \\ &+ \beta_2.Informed\ Push\ Model_{rt} + \varepsilon_{hjt}. \end{aligned} \quad (2)$$

This specification is based on a logistic regression model, where h , j , r , and t denote health worker h at facility j located in region r at time t and the unit of analysis is the facility–health worker. To alleviate concerns of bias from confounding variables, we adopt a two-pronged approach: (i) matching techniques (see Online Table B.1b for balance diagnostics) and (ii) inclusion of a proxy variable to capture the potential endogeneity of rollout timing (i.e., *Catch. Pop. Proxy*).

Subsequently, to evaluate the impact of the informed push model on client satisfaction, we have access to only two rounds of survey data (i.e., 2012–2013 and 2015 client exit interview questionnaires). Given this limitation, we lack sufficient variation in rollout timing across regions to estimate a DiD model specification similar to Equation (2). We, therefore, identify the effect of the informed push model on client satisfaction using a prepost comparison design, which has been previously used to evaluate the impact of public health initiatives (see Rowe et al. 2018). We utilize the following logistic regression specification in our analysis:

$$\begin{aligned} & \text{Ln} \left[\frac{\Pr(\text{Client Satisfaction}_{cjt} = 1)}{1 - \Pr(\text{Client Satisfaction}_{cjt} = 1)} \right] \\ &= \beta_0 + \lambda.X_{CL} + \text{Catch.Pop.Proxy}_{rt} \\ &+ \beta_3.Informed\ Push\ Model_t + \varepsilon_{cjt}, \end{aligned} \quad (3)$$

where c , j , r , and t denote client c at facility j located in region r at time t , and the unit of analysis is facility–client. To evaluate these relationships of interest, we apply each of the specifications in Equations (2) and (3) to subgroups of health facilities categorized based on a health facility's *LMIS Practices* and access to *Road Infrastructure*. We use the same two-pronged approach

discussed earlier to alleviate endogeneity concerns (see Online Table B.1c for CEM balance diagnostics.)

4.3.3. Additional Client-Related Outcomes. In addition to directly estimating the effect of the transition to the informed push model on client satisfaction as part of RQ2, we evaluate the impact on alternative public health outcomes (e.g., unintended pregnancies, maternal and newborn mortalities.) We first note that directly estimating the impact of the transition on such public health outcomes is fraught with challenges (e.g., because of the presence of unobserved confounding variables, concerns related to data quality and availability.) To circumvent these challenges, we draw upon existing literature (Reproductive Health Supplies Coalition 2022) to leverage the first order effects of the transition to the informed push model, that is, its impact on contraceptive stock-outs, as the basis for evaluating the public health impacts of the transition. In particular, we utilize the reducing stock-outs impact calculator (RSIC) tool developed by the Reproductive Health Supplies Coalition to calculate how the reductions in stock-outs discussed in Section 4.2 translate into improvements in the following public health outcomes: (i) contraceptive prevalence rate, (ii) number of unintended pregnancies averted, and (iii) number of maternal and newborn deaths averted. The RSIC tool is built based on estimates from prior studies (e.g., Ross and Stover 2013) that show the contraceptive prevalence rate decreases by 6.50 p.p. when health facilities reduce the number of contraceptive methods they offer by one. Based on the assumption that the stock-out of a contraceptive method equates to the partial removal of that method from the shelf, the RSIC tool quantifies how changes in stock-outs translate into changes in contraceptive prevalence rate. The tool then utilizes estimates from previous research (see Darroch 2017) to further translate the changes in contraceptive prevalence rate into changes in the number of unintended pregnancies and maternal and newborn deaths (for details regarding the assumptions and limitations of this tool, see Online Appendix A.1).

4.4. Evaluating the Economic Implications of the Transition to the Informed Push Model (RQ3)

4.4.1. Cost Estimation Approach. To the best of our knowledge, there are no data sources or published studies that report the facility-level cost of transitioning to the informed push model in Senegal. To circumvent this challenge, we collected data regarding the cost of implementing the informed push model at the upstream levels of the reproductive health supply chain in Senegal (i.e., national, regional, and district levels). Leveraging this information, we utilized the following cost-approximation approach proposed by Lynch et al. (2020) to allocate the costs incurred at the upstream levels

to the downstream health facilities:

$$\begin{aligned} & \text{Cost Allocated to a Health Facility} \\ &= \frac{\left(\frac{\text{National Cost}}{\text{No. of Regions}} + \text{Regional Cost} \right) + \text{District Cost}}{\text{No. of Districts in a Specific Region}} \\ &= \frac{\text{No. of Health Facilities in a Specific District}}{\text{Health Facility Cost}}, \end{aligned} \quad (4)$$

where the facility-level costs of operating the informed push model are calculated by progressively allocating upstream costs to the next downstream level in a proportional fashion. We follow a similar approach to calculate the facility-level costs of operating the pull distribution model. The difference between the two values provides an estimate of the cost of transitioning to the informed push model at each health facility. We provide more detailed information on this cost estimation strategy in Online Appendix A.2. Further, we evaluate the robustness of our cost estimates by recalculating them using an alternative costing approach outlined in Burns et al. (1985). This approach, explained in greater detail in Online Appendix A.3, takes into account additional factors, such as vehicle routing and facility density.

4.4.2. Cost–Benefit Ratios. The next step is to calculate the cost–benefit ratio of the transition from the pull model to the informed push model. In our analysis, we define the benefit for each health facility as the p.p. change in the likelihood of stock-outs because of the transition, which we estimate from our regression specification presented in Equation (1). The facility-level cost–benefit ratio is calculated as follows:

$$\begin{aligned} & \text{Cost-Benefit Ratio at a Health Facility} \\ &= \frac{\text{Transition Cost}}{\text{Percentage Point Change in Stock-Out}}. \end{aligned} \quad (5)$$

Using Equation (5), we calculate the average cost–benefit ratios for different types of health facilities classified based on the maturity of their LMIS practices and access to road infrastructure. We then use these cost–benefit ratios to address our third research question (RQ3), which focuses on identifying the specific types of facilities that should be prioritized for transition to the informed push model.

5. Results

5.1. Operational Implications of the Transition to the Informed Push Model (RQ1)

To address our first research question (RQ1), we discuss the estimation results pertaining to the effects of the informed push model on contraceptive stock-outs. First, we investigate how a health facility's LMIS practices moderates the relationship between the distribution model and contraceptive stock-outs. The results in Table 1, columns (2) and (3), show that a transition to the

informed push model leads to a larger reduction in stock-outs for facilities with less mature LMIS practices (column (2): $\beta = -2.96$, $p < 0.01$) relative to the ones with more mature practices (column (3): $\beta = -1.45$, $p < 0.01$). In terms of the magnitude of effect sizes, the transition reduces the probability of stock-outs by 25.14 p.p. at facilities with less mature LMIS practices compared with a 12.53 p.p. reduction at facilities with more mature practices ($\chi^2 = 10.44$, $p < 0.01$).

Next, we interpret the results of how a health facility's access to road infrastructure moderates the benefits of transitioning to the informed push model. Columns (4) and (5) in Table 1 indicate that the transition leads to a larger reduction in stock-outs for facilities with access to less developed road infrastructure (column (4): $\beta = -2.33$, $p < 0.01$) compared with those with access to more developed road infrastructure (column (5): $\beta = -1.35$, $p < 0.01$). Specifically, the transition reduces the probability of stock-outs by 20.39 p.p. at facilities with less developed road infrastructure compared with 11.40 p.p. at facilities with more developed infrastructure ($\chi^2 = 4.48$, $p < 0.05$).

Finally, we investigate whether and to what extent the potential benefits of transitioning to the informed push model are jointly moderated by a facility's LMIS practices and access to road infrastructure. To this end, we estimate Equation (1) across the subgroups of facilities with less mature LMIS practices and less developed road infrastructure (i.e., severely disadvantaged) versus those with more mature LMIS practices and more developed road infrastructure (i.e., nondisadvantaged). The results are presented in columns (6) and (7) of Table 1. We observe that the transition is associated with a larger reduction in stock-outs for facilities that are severely disadvantaged (column (6): $\beta = -3.71$, $p < 0.01$) compared with those in the nondisadvantaged group (column (7): $\beta = -1.18$, $p < 0.01$). Specifically, the transition leads to a 30.00 p.p. reduction in stock-outs at severely disadvantaged facilities, whereas the benefits are roughly one third (9.60 p.p.) for those in the nondisadvantaged group ($\chi^2 = 16.58$, $p < 0.01$). Taken together, our results suggest that the operational benefits of the transition in terms of reductions in stock-outs are larger for health facilities with less mature LMIS practices and/or access to less developed road infrastructure, thereby addressing RQ1.

5.2. Public Health Implications of the Transition to the Informed Push Model (RQ2)

5.2.1. Frontline Health Worker Workload. To address our second research question (RQ2), we begin by interpreting the results in Table 2, which summarizes the impact of the transition to the informed push model on the workload of frontline health workers. From columns (2) and (3) in the table, we observe that the informed

Table 1. Triple Differences Estimation on Coarsened Exact Matched Samples: Effects of Transition from the Pull Model to the Informed Push Model on *Stock-Out*

Dependent variable: <i>Stock-Out</i>	All (1)	LMIS only		Road Infrastructure Only		LMIS and Road Infrastructure	
		Less mature (2)	More mature (3)	Less developed (4)	More developed (5)	Less mature and less developed (6)	More mature and more developed (7)
<i>Informed Push Model</i> ×	–1.80*** (0.26)	–2.96*** (0.51)	–1.45*** (0.24)	–2.33*** (0.30)	–1.35*** (0.36)	–3.71*** (0.58)	–1.18*** (0.36)
<i>Contraceptive</i>							
<i>Marginal Effect</i> , p.p.	–15.86	–25.14	–12.53	–20.39	–11.40	–30.00	–9.60
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Catch. Pop. Proxy</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Commodity variety fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Stock-out base rate	19.12	27.06	16.62	22.81	15.79	31.60	14.72
Pseudo R^2	0.16	0.20	0.18	0.19	0.17	0.29	0.22
Observations	8,105	2,228	5,877	4,052	4,053	1,215	3,043

Notes. Unit of analysis is facility-health commodity. Robust standard errors clustered at the district level in parentheses. In addition to the listed covariates, the model in column (1) also controls for *LMIS Practices* and *Road Infrastructure*. Observations were matched based on the following variables: *Facility Type*, *Number of Full-time Workers*, and *Number of Part-time Workers*.

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

push model leads to a reduction in workload at facilities with less mature LMIS practices (column (2): $\beta = -1.40$, $p < 0.01$). The effect is, however, statistically insignificant at facilities with more mature LMIS practices (column (3): $\beta = 0.05$, n.s.). Next, the results in columns (4) and (5)

of Table 2 show that the benefits of the transition are statistically insignificant at facilities with both less and more developed road infrastructure (column (4): $\beta = -0.54$, n.s., and column (5): $\beta = -0.12$, n.s.). Finally, we observe from columns (6) and (7) of the table that the informed

Table 2. Difference-in-Differences Estimation on Coarsened Exact Matched Samples: Effects of Transition from the Pull Model to the Informed Push Model on *Frontline Health Worker Workload*

Dependent variable: <i>Frontline Health Worker Workload</i>	All (1)	LMIS only		Road infrastructure only		LMIS and road infrastructure	
		Less mature (2)	More mature (3)	Less developed (4)	More developed (5)	Less mature and less developed (6)	More mature and more developed (7)
<i>Informed Push Model</i>	–0.22 (0.21)	–1.40*** (0.44)	0.05 (0.22)	–0.54 (0.41)	–0.12 (0.25)	–4.08*** (0.69)	0.19 (0.26)
<i>Marginal Effect</i> , p.p.	–3.24	–18.52	0.82	–7.90	–1.74	–62.17	2.85
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Catch. Pop. Proxy</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Commodity variety fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.10	0.17	0.10	0.11	0.12	0.18	0.13
Observations	4,504	1,193	3,271	1,706	2,761	485	2,081

Notes. Unit of analysis is facility-health worker. Robust standard errors clustered at the district level in parentheses. In addition to the listed covariates, the model in column (1) also controls for *LMIS Practices* and *Road Infrastructure*. Observations were matched based on the following variables: *Facility Type*, *Number of Full-time Workers*, and *Number of Part-time Workers*. Results are robust to the parallel trends relaxation approach proposed by Freyaldenhoven et al. (2019).

* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

push model leads to a larger reduction in workload at health facilities with less mature LMIS practices and less developed road infrastructure (column (6): $\beta = -4.08$, $p < 0.01$), relative to facilities with more mature LMIS practices and more developed road infrastructure (column (7): $\beta = 0.19$, n.s.). Of note, for facilities with less mature LMIS practices and less developed road infrastructure, the coefficient equates to a 62.17 p.p. decrease in the probability that heavy workload compromises a frontline health worker's ability to deliver clinical services.

5.2.2. Client Satisfaction. Next, we discuss how the transition impacts client satisfaction. From Table 3, we find that the transition to the informed push model leads to (i) a larger increase in client satisfaction at facilities with less mature LMIS practices compared with those with more mature LMIS practices (column (2): $\beta = 4.33$, $p < 0.01$; column (3): $\beta = 2.28$, $p < 0.01$; $\chi^2 = 1.65$, n.s.), (ii) a larger increase in client satisfaction at facilities with less developed road infrastructure compared with those with more developed road infrastructure (column (4): $\beta = 2.98$, $p < 0.01$; column (5): $\beta = 1.64$, $p < 0.01$; $\chi^2 = 1.89$, n.s.), and (iii) a larger increase in client satisfaction at facilities with less mature LMIS practices and less developed road infrastructure (column (6): $\beta = 6.41$, $p < 0.01$) compared with those with more mature LMIS practices and more developed road infrastructure (column (7): $\beta = 2.34$, $p < 0.01$; $\chi^2 = 3.50$, $p < 0.10$). Importantly, we find that, across all subgroups of facilities, the largest effect sizes are observed at severely disadvantaged facilities (column (6)), with a 28.89 p.p. increase in the probability that a client leaves the facility with no complaints regarding the availability of contraceptives.

5.2.3. Additional Client-Related Outcomes. Finally, we estimate the effect of the informed push model on other client outcomes, including contraceptive prevalence rate, unintended pregnancies, and maternal and newborn deaths. The relevant results are presented in Table 4. For example, we infer from column (6) in Table 4 that facilities with less mature LMIS practices and less developed road infrastructure benefit by far the most from a transition to the informed push model, experiencing a 17.55 p.p. increase in contraceptive prevalence rate, 6,247 unintended pregnancies averted, and 145 maternal and newborn mortalities averted per 100,000 women of reproductive age using contraception. In contrast, column (7) in Table 4 suggests that the benefits of the transition are noticeably smaller for facilities with more mature LMIS practices and more developed road infrastructure, translating into a 5.62 p.p. increase in contraceptive prevalence rate, 2,000 unintended pregnancies averted, and 46 mortalities averted. Taken together, we conclude that the public health benefits of the transition are substantially larger for health facilities with less mature LMIS practices and/or access to less developed road infrastructure, thereby addressing RQ2.

5.3. Economic Implications of the Transition to the Informed Push Model (RQ3)

To address our third research question (RQ3), we first summarize the results by highlighting the operational benefits (i.e., reductions in stock-outs) of the transition to the informed push model in Figure 3(a). From the figure, we observe that (i) severely disadvantaged health facilities (i.e., less mature LMIS and less developed road infrastructure) obtain the largest benefits from a transition to the informed push model; (ii) moderately disadvantaged

Table 3. Prepost Comparison Estimation on Coarsened Exact Matched Samples: Effects of Transition from the Pull Model to the Informed Push Model on *Client Satisfaction*

Dependent variable: <i>Client Satisfaction</i>	All (1)	LMIS only		Road infrastructure only		LMIS and road infrastructure	
		Less mature (2)	More mature (3)	Less developed (4)	More developed (5)	Less mature and less developed (6)	More mature and more developed (7)
<i>Informed Push Model</i>	2.14*** (0.50)	4.33*** (1.50)	2.28*** (0.54)	2.98*** (0.88)	1.64*** (0.42)	6.41*** (2.13)	2.34*** (0.45)
<i>Marginal Effect, p.p.</i>	14.47	19.00	15.35	20.92	9.93	28.89	14.09
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day-of-week fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Catch. Pop. Proxy</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Commodity variety fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.29	0.45	0.34	0.43	0.24	0.62	0.32
Observations	1,491	311	1,154	576	831	128	666

Notes. Unit of analysis is facility–client. Robust standard errors clustered at the district level in parentheses. In addition to the listed covariates, the model in column (1) also controls for *LMIS Practices* and *Road Infrastructure*. Observations were matched based on the following variables: *Facility Type*, *Number of Full-time Workers*, and *Number of Part-time Workers*.
* $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 4. Reducing Stock-outs Impact Calculator Tool Estimates: Effects of Transition from the Pull Model to the Informed Push Model on Additional Client Outcomes

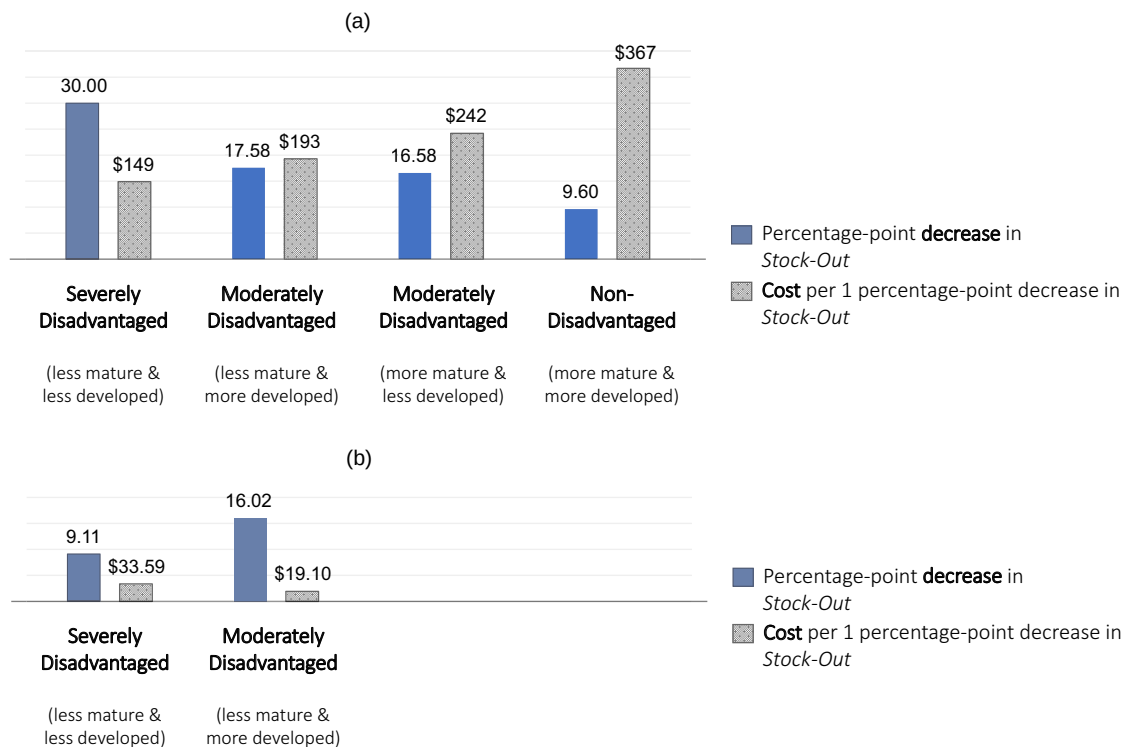
Additional Client Outcomes	All (1)	LMIS only		Road infrastructure only		LMIS and road infrastructure	
		Less mature (2)	More mature (3)	Less developed (4)	More developed (5)	Less mature and less developed (6)	More mature and more developed (7)
Contraceptive prevalence rate increase, p.p.	9.29	14.71	7.33	11.93	6.67	17.55	5.62
Number of pregnancies averted (per 100,000 women of reproductive age)	3,305	5,234	2,610	4,245	2,374	6,247	2,000
Number of maternal and newborn mortalities averted (per 100,000 women of reproductive age)	76	121	60	98	55	145	46

Notes. Estimates are based on the Reproductive Health Supply Coalition's RSIC tool. See Online Appendix A.1 for details regarding the underlying assumptions and limitations of this tool.

facilities (i.e., either less mature LMIS or less developed road infrastructure, but not both) benefit relatively less; and (iii) those that are nondisadvantaged (i.e., more mature LMIS and more developed road infrastructure) experience the smallest benefits.

Next, the cost-benefit ratios reported in Figure 3(a) (see Equation (5) for the estimation approach) indicate

that transitioning to the informed push model is most cost-efficient for the severely disadvantaged facilities (average cost of \$149 per 1 p.p. reduction in stock-outs), followed by the two groups of moderately disadvantaged facilities (average cost of \$193 and \$242 per 1 p.p. reduction in stock-outs). For facilities that are nondisadvantaged, the cost-benefit ratio is significantly larger

Figure 3. (Color online) Benefits and Cost-Benefit Ratios for *Stock-Out*

Notes. (a) Transition from the Pull Model to the Informed Push Model. (b) Upgrading LMIS Practices from Less Mature to More Mature. All benefit values are statistically significant at $p < 0.05$. All cost numbers are reported in U.S. dollars.

(average cost of \$367 for a 1 p.p. reduction in stock-outs). Hence, severely disadvantaged facilities not only reap the largest stock-out reduction from the transition but also achieve this benefit in the most cost-efficient manner.

Beyond stock-out reductions, we find that severely disadvantaged health facilities lead the way in terms of realized benefits across a spectrum of alternative outcomes, achieving these results more cost-efficiently than other facilities (see Figure 4). These outcomes include (a) frontline health worker workload, (b) client satisfaction, (c) contraceptive prevalence rate, (d) unintended pregnancies, and (e) maternal and newborn mortalities. For example, achieving a 1 p.p. increase in contraceptive prevalence rate incurs an average cost of \$169 for the severely disadvantaged facilities, whereas the costs increase substantially to between \$277 and \$309 for the moderately disadvantaged facilities and \$476 for the nondisadvantaged facilities (Figure 4(c)). Similarly, averting one death in the catchment population of severely disadvantaged health facilities costs about \$20.49, whereas achieving the same outcome incurs a significantly larger cost of between \$33.59 and \$37.55 for the moderately disadvantaged facilities and \$57.85 for facilities that are nondisadvantaged (Figure 4(e)). We observe broadly consistent patterns across other outcomes in Figure 4, (b) and (d).

Although our study is focused on evaluating the impact of the transition from pull model to the informed push model, alternative initiatives with different cost structures exist to improve the availability of health commodities. One such initiative is investing to enhance the maturity of a facility's LMIS practices. Based on estimates reported in the prior literature (e.g., McElwee 2015, U.S. Agency for International Development 2019), the cost of upgrading a facility's LMIS practices from less mature to more mature is roughly \$306 per health facility (see Online Appendix A.4 for details). Using our specification in Equation (A1) in Online Appendix A.4, we estimate the reduction in stock-out because of LMIS upgrades at facilities with less developed road infrastructure to be 9.11 p.p. (see Figure 3(b)). The corresponding stock-out reduction at health facilities with more developed road infrastructure is 16.02 p.p. Together, these estimates yield a set of cost-benefit ratios associated with enhancing the maturity of a facility's LMIS practices, which are reported in Figure 3(b).

Comparing the values across Figure 3, (a) and (b), for the severely disadvantaged health facilities, we find that achieving a 1 p.p. reduction in stock-outs involves an average cost of \$149 for the informed push model and \$33.59 for upgrading the LMIS practices. This result indicates that upgrading the LMIS practices might be more cost-efficient than transitioning to the informed push model for such facilities. However, delving deeper into the benefit values for these health facilities in Figure 3(a)

reveals nuanced insights. Specifically, although it is more cost-efficient, the maximum benefits from upgrading the LMIS practices are less than one third of those achieved by the informed push model for severely disadvantaged health facilities (9.11 and 30.00 p.p. decrease in stock-outs, respectively). In essence, the informed push model costs more, but exhibits a leapfrogging effect (see Jeon et al. 2023) that elevates these health facilities to a substantially higher level of performance than what they can attain through upgrading the LMIS practices. Hence, when funding permits, it is advisable to transition these health facilities to the informed push model.

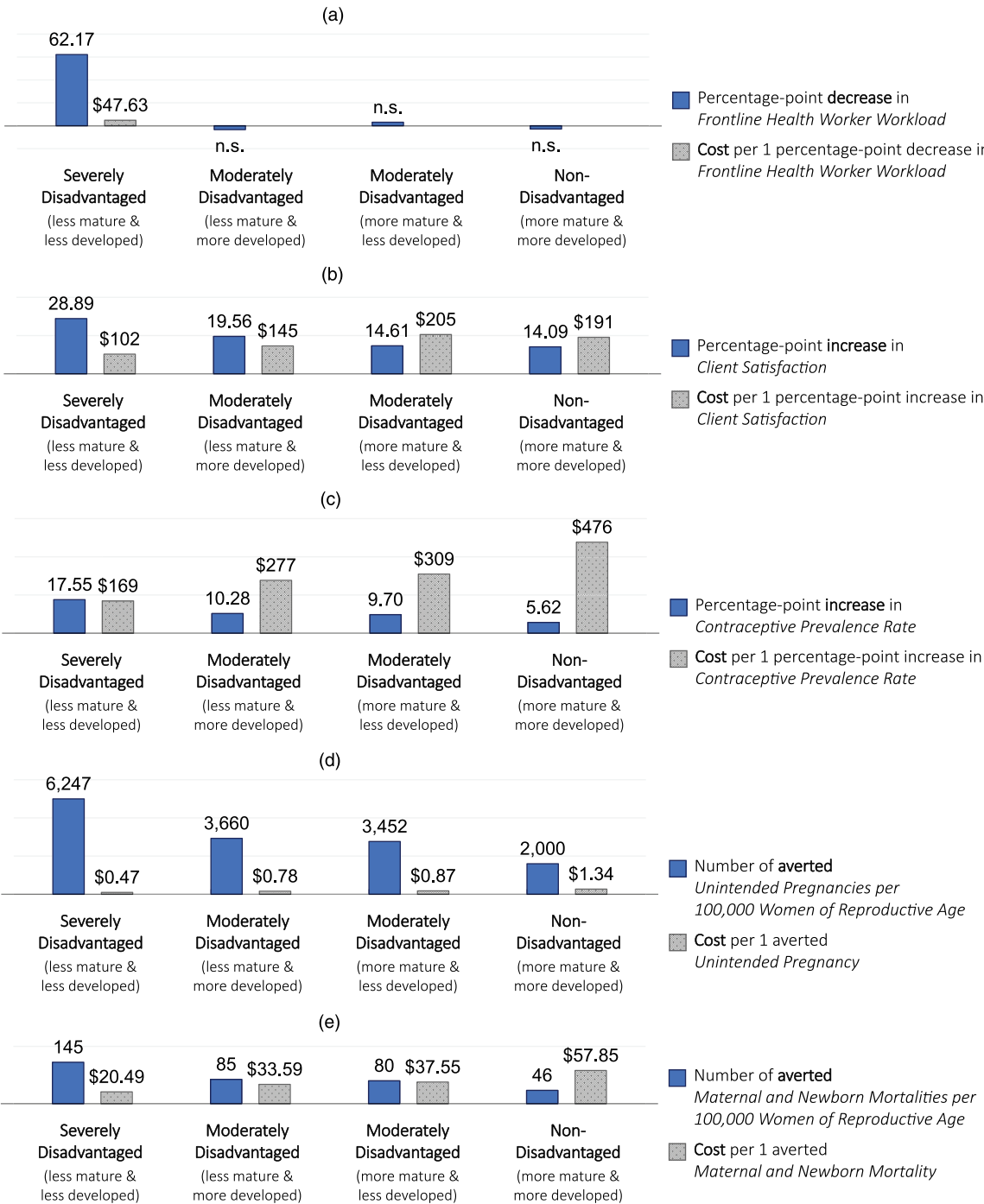
On the other hand, a 1 p.p. reduction in stock-outs for the moderately disadvantaged health facilities carries an average cost of \$193 for the informed push model and \$19.10 for upgrading the LMIS (see Figure 3, (a) and (b)). This stark difference in costs arises despite both interventions yielding relatively comparable benefits (i.e., a decrease of 17.58 p.p. in stock-outs for implementing the informed push model compared with 16.02 p.p. for the latter). Consequently, in the case of moderately disadvantaged health facilities, upgrading the LMIS practices emerges as a significantly more cost-efficient strategy than a transition to the informed push model.

In sum, the informed push model proves most advantageous for the severely disadvantaged facilities. In contrast, moderately disadvantaged facilities are better suited for LMIS upgrades, whereas the nondisadvantaged should not be prioritized for resource allocation. Taken together, these findings address RQ3.

6. Robustness Checks

We conducted several robustness checks to ensure the reliability and validity of the estimates from our analyses. Our first check is the Goodman-Bacon decomposition (Online Appendix A.5), which evaluates the potential for bias in staggered designs (Goodman-Bacon 2021). The results show that the treatment effects are not likely to be biased because of negative weighting issues. The second check is the placebo falsification test (Online Appendix A.6), which was conducted to assess bias from unobserved confounders. The test involves estimating DiD models using an alternative outcome variable that was unaffected by the treatment. The results of the test show that the placebo rollout of the informed push model to basic health supplies did not lead to any consistent and statistically significant changes in the availability of these products. This suggests that the main findings are not likely to be driven by facility-level time-invariant unobserved confounders. The third check is a two-staged least squares endogeneity-corrected event study (Online Appendix A.7), which relaxes the parallel trends assumption (Freyaldenhoven et al. 2019). The results show that the estimates are robust to potential violations of the parallel trends assumption, confirming

Figure 4. (Color online) Benefits and Cost-Benefit Ratios for Public Health Outcomes



Notes. (a) Frontline health worker workload. (b) Client satisfaction. (c) Contraceptive prevalence rate. (d) Unintended pregnancies averted. (e) Maternal and newborn mortalities averted: transition from the pull model to the informed push model. All estimated benefit values in (a) and (b) are statistically significant at $p < 0.05$ except for those marked as “n.s.” (not statistically significant). All cost numbers are reported in U.S. dollars. The number of women of reproductive age (15–49 years old) in Senegal is 3,270,712 and 4,394,262 based on estimates from 2013 and 2023 from the Senegalese National Agency of Statistics and Demography, respectively.

the validity of the estimates. The fourth check is the instrument-free Gaussian Copula correction (Online Appendix A.8), which addresses concerns related to the endogeneity of the rollout timing. The results obtained

using this approach are consistent with the main model, suggesting that the coefficients are not biased because of the potential endogeneity of the rollout timing. The fifth and final check involves the application of propensity

score matching in lieu of CEM and incorporating additional variables in the matching process, leading to fully consistent results.¹⁵ Collectively, these checks provide strong evidence for the robustness of the estimated coefficients.

7. Conclusion

7.1. Summary and Implications for Managing Public Health Supply Chains

The limited access to reproductive health commodities in LDCs has profound implications for women. These repercussions span from unintended pregnancies and lost career aspirations to mental health concerns (e.g., distress, depression) and physical health complications (e.g., injuries, infections, and even deaths). Such limited access can be partly attributed to ineffective supply chains for distributing health commodities in LDCs. A prime example is the commonly prevalent pull distribution model in LDCs, which requires frontline health workers to be in charge of inventory management on top of their clinical responsibilities. This setup increases the risk of supply chain failures, such as stock-outs. In response, several LDCs have adopted an alternative distribution approach known as the informed push distribution model. This model shifts inventory management to external logistics providers, freeing health workers from this added responsibility. Leveraging novel field data collected from multiple entities across Senegal—a country that redesigned its contraceptive distribution system—we evaluate the operational (RQ1), public health (RQ2), and economic implications (RQ3) of the transition from the pull model to the informed push model. Our findings have actionable implications for the effective and efficient allocation of limited resources to expand access to reproductive health commodities in LDCs.

First, we find that the transition to the informed push model leads to significant benefits across a spectrum of operational and public health outcomes. These benefits are particularly salient in the case of severely disadvantaged health facilities that face significant obstacles with respect to effective inventory management. Our findings show that these health facilities experience the largest reduction in stock-outs, equivalent to 30.00 p.p. In contrast, the benefits are more modest for moderately disadvantaged facilities (17.58 and 16.58 p.p. decrease in stock-outs), and facilities that are nondisadvantaged experience even less benefits from the transition (9.60 p.p. reduction in stock-outs).

Second, our findings indicate that severely disadvantaged health facilities experience the largest reduction in the workload of frontline health workers (62.17 p.p. decrease). The reduction in workload is likely to yield significant public health benefits by enabling the frontline health workers to allocate more time to patient care.

This is affirmed by field interviews in which health workers noted that “they now have an easier time fulfilling their duties” (McElwee 2015, p. 28). Further, by alleviating the workload burden, the informed push model is expected to reduce job dissatisfaction, burnout, and turnover rates among frontline health workers. This is especially relevant in the context of Senegal and LDCs more broadly, where health workers are reported to be experiencing severe signs of burnout (Rouleau et al. 2012). Our finding also complements prior studies on better supporting the frontline health workers in performing their key tasks, whether through the provision of training (Rowe et al. 2018), leveraging information technologies (U.S. Agency for International Development 2019), or the use of external logistics providers as seen with the informed push model.

Third, severely disadvantaged facilities experience the largest increase in client satisfaction, whereas the benefits are more modest for moderately disadvantaged and nondisadvantaged facilities. Such improvements in client satisfaction could subsequently translate into increased contraceptive usage, reduced likelihood of method discontinuation, heightened trust in the public health delivery system, and ultimately improved public health outcomes in LDCs.

Fourth, our findings show that the catchment population of severely disadvantaged health facilities stand to gain the most from the transition in terms of improvements in public health outcomes. For example, we find that the informed push model would help to avert 6,247 unintended pregnancies and 145 maternal and newborn mortality cases per 100,000 women of reproductive age within this population annually. These estimates have meaningful implications for the health and well-being of communities in Senegal. Specifically, our numerical calculations show that the reduction in maternal and newborn mortalities because of the informed push model would translate into a 32% decline in mortality rates for the populations served by these health facilities in Senegal. This effect is comparable to the roughly 27% decrease in death rates resulting from demand-side public awareness campaigns focused on promoting contraceptive usage.¹⁶ This underscores the significant potential of supply-side approaches in improving health outcomes in LDCs.

Finally, our cost-benefit analyses reveal that, for severely disadvantaged health facilities, the informed push model is a more costly intervention than upgrading LMIS practices but, in turn, exhibits a leapfrogging effect that delivers substantially larger benefits (Jeon et al. 2023). Hence, when funding permits, it is advisable to transition these health facilities to the informed push model. In contrast, for moderately disadvantaged facilities, upgrading the LMIS practices emerges as a significantly more cost-efficient strategy to reduce stock-outs than a transition to the informed push model. Finally,

our results indicate that the nondisadvantaged health facilities should not be prioritized for funding allocation to transition to the informed push model. Together, our findings serve to inform the effective and cost-efficient design of last-mile distribution strategies to enhance health commodity access in LDCs.

7.2. Limitations and Future Research Directions

Our study has some limitations that serve as avenues for future research. One such limitation is related to our dependent variable, which captures stock-outs on the day of the survey visit. Although this measure is well-established in the literature (see Choi and Ametepi 2013), there is potential for underestimating or overestimating stock-outs based on the survey data at a facility. However, the stratified random sampling nature of the SPA surveys¹⁷ and the inclusion of day-of-week fixed effects in all models should help alleviate such concerns. Additionally, our examination of the informed push model's impact on client satisfaction relies on a prepost comparison design because of limited data availability. This presents an opportunity for future research to delve deeper into this relationship with more comprehensive data.

Another limitation is that the rollout of the informed push model involved a change in the payment structure, which we are not able to fully disentangle from the effects of changes in the logistics cycle activities. However, Krug et al. (2020) argue that, whereas changes in the logistics activities were a significant contributor to reducing the stock-out risks, the influence of the payment structure remains less clear. Further, we believe that the observed reduction in workload because of the informed push model is likely to be exclusively driven by changes to the logistic cycle activities rather than the payment structure. This conjecture is supported by Daff et al. (2014, p. 249), who provide qualitative evidence that “the informed push model has freed up health workers to spend more time providing care to patients.” Nevertheless, examining the impact of payment structure in this context remains an open avenue for future research.

Finally, we execute our empirical analysis in the context of reproductive health supply chains in Senegal, West Africa. Future research could replicate our findings by focusing on alternative geographical contexts. Notwithstanding these limitations, our paper uncovers nuanced insights on how the type of distribution model used for delivering health commodities in public health supply chains impacts the last-mile availability of reproductive health supplies.

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Endnotes

¹ This is defined as the proportion of women who intend to prevent or delay a pregnancy but lack access to suitable methods of contraception.

² It is estimated that the operating costs associated with the informed push distribution of contraceptive methods in Senegal is equivalent to 11% of the annual national spending on contraceptive procurement (Daff et al. 2014).

³ According to the World Health Organization, maternal mortality is defined as the death of a woman while pregnant or within 42 days of termination of pregnancy from any cause related to or aggravated by the pregnancy or its management but not from accidental or incidental causes. Newborn or neonatal mortality is defined as the death of a newborn within 28 days of birth.

⁴ For the 2012–2013 survey, the data collection extended from 2012 into the following year.

⁵ The SPA's inventory questionnaire collected data from 289, 290, and 286 health facilities offering at least one contraceptive method during the survey rounds of 2012–2013, 2014, and 2015, respectively. After strictly matching the GPS coordinates of facilities across the three surveys, we identified 117 repeated facilities across the survey rounds of 2012–2013 and 2015. No repeated facilities were found across the survey rounds of 2012–2013 and 2014 and 2014 and 2015. Therefore, our data primarily consists of nationally representative repeated cross-sections of health facilities.

⁶ The client exit interview data on contraceptive availability were not available for the 2014 SPA survey.

⁷ The DHS's women and household questionnaires collected data from 8,636, 8,488, and 8,851 women across the survey rounds of 2012–2013, 2014, and 2015, respectively.

⁸ These cases were rare, comprising less than 2% of the observations.

⁹ Online Appendix A is available for download from the journal website.

¹⁰ Our results remain consistent when we apply no lag, a one-month lag, and a three-month lag.

¹¹ In addition, deworming medications exhibit comparable characteristics to contraceptives in terms of demand predictability and storage requirements (see Yadav et al. 2014), justifying their selection as placebo units. We also note that, during the later stages of the rollout of the informed push model (i.e., after 2016), additional health commodities beyond contraceptives were included in the delivery program (e.g., HIV tests). We do not evaluate the impact of this expanded rollout of the informed push model, which is called *Yeksi Naa* (Krug et al. 2020).

¹² These methods are male condoms, female condoms, combined pills, progestin-only pills, combined injectables, progestin-only injectables, intrauterine devices, implants, emergency contraceptives, and cycle beads.

¹³ These medications are albendazole and mebendazole.

¹⁴ Online Appendix B is available for download through the following link: <https://github.com/karimamiri/Senegal/blob/main/AppendixB.pdf>.

¹⁵ Results are not shown to save space but are available upon request.

¹⁶ The calculations take into account birth rate, maternal mortality rate, and newborn mortality rate in Senegal. Based on 2012 estimates from the World Bank, the values are estimated to be 3,700 per 100,000 people, 394 per 100,000 live births, and 2,600 per 100,000 live births, respectively. Further, the total number of women of reproductive age (15–49 years old) and the total population of Senegal are estimated to be 3,270,712 and 13,508,715, respectively, based on 2013 estimates from the Senegalese National Agency of Statistics and Demography. Next, the total number of maternal and newborn mortalities in Senegal is estimated to be 458 per 100,000 women of reproductive age. The estimated benefits of public awareness campaigns are based on the study by Hutchinson and Meekers (2012), who find an average increase of 15.15 p.p. in contraceptive prevalence rate because of these campaigns.

¹⁷ Incorporating the survey sampling weights into the regression models leads to fully consistent results.

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